

2022-Ph-H1-Q6

Section: Our Dynamic Universe

Topic: Energy and Power

Summary:

Water flows at a rate of $1.0 \times 10^6 \text{ kg/s}$ over a 120 m drop (Victoria Falls). Find the total **power** delivered by the falling water.

Solution:

Potential energy per second (power):

$$P = \dot{m}gh = 1.0 \times 10^6 \times 9.8 \times 120 = 1.18 \times 10^9 \text{ W} \approx 1.2 \times 10^9 \text{ W}.$$

Answer: B. $1.2 \times 10^9 \text{ W}$.

Guidance for Students:

- Power is energy per second; here, gravitational potential energy is converted each second.

Revision Tips:

- Use $P = \dot{m}gh$ for falling mass flow.
- Ensure mass flow rate is in kg/s, height in metres.